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PAASCU LEVEL II: Higher Education
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PROJECT PROGRESS REPORT

Smart Security with Arduino: Automated Locking System

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I. PROJECT OVERVIEW

The RFID Vault Access System using Arduino is designed to provide a reliable and secure access control system for vaults, safe rooms, or restricted areas. The system uses RFID tags or cards to authenticate users and grant or deny access, while logging all access attempts and sending notifications for unauthorized entries. The purpose of this project is to enhance security, prevent unauthorized access, and provide a convenient and efficient way to manage access control without relying on traditional keys. By incorporating an alert system for unauthorized access attempts or system tampering, the project ensures both safety and convenience. In today's fast-paced world, securing valuable assets and sensitive information is crucial, as traditional lock-and-key systems can be vulnerable to unauthorized entry, lost keys, and tampering. Our project aims to offer a secure, efficient, and reliable solution for controlling access to restricted areas or assets using RFID technology, making it an ideal system for businesses, individuals, and organizations.

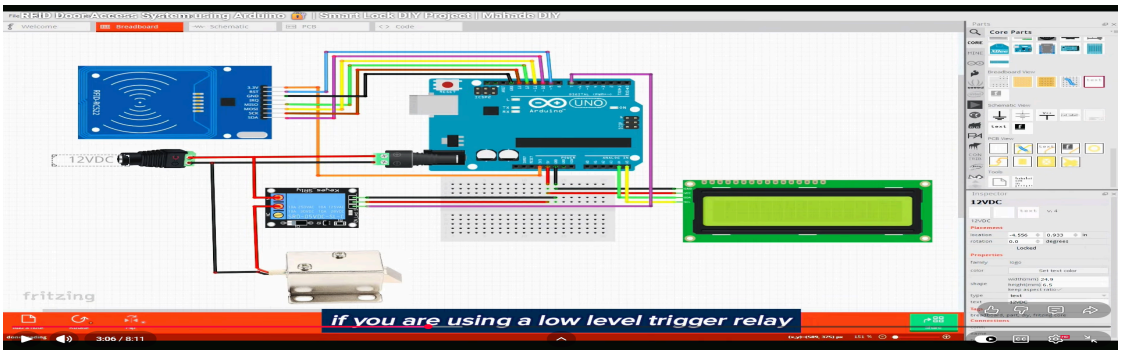
II. Progress Timeline

Phase 1 – Conceptual Proposal

In the first phase, January 9th, the team worked on defining the main idea and goals of the Arduino project. We discussed different project ideas and checked if they were possible based on the parts we had, the time available, and the skills of each group member. By the end of this phase, we had agreed on a project concept, created a clear plan for the next steps, and organized how each member would contribute to the work.

Phase 2 – System Simulation

On January 12th , the team used Tinkercad to simulate the RFID system and test the necessary components to a virtual Arduino board to see if the system design would function as planned. The simulation helped us identify potential issues with connections and component placement before assembling the actual hardware. This process allowed the team to plan the assembly more efficiently and reduce errors during physical testing.



Phase 3 – Component Procurement

January 20th, the team focused on gathering all the parts needed for the Arduino project. We identified the required components and purchased them online, including:

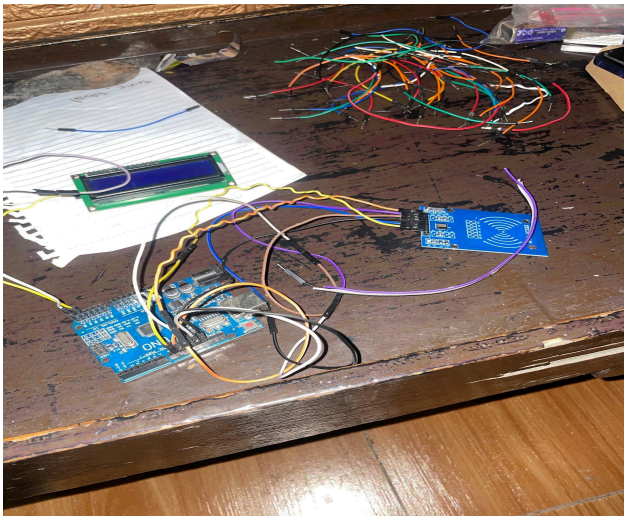
	Arduino Uno	The main microcontroller board that controls the entire system, reads inputs from sensors, and sends signals to other components.
	RFID RC522 Module + RFID Card / Key Tag	Reads RFID cards or tags to identify authorized users and send that information to the Arduino for access control.
	16x2 LCD Display with I2C Adapter	Shows messages or status updates, such as “Access Granted” or “Access Denied,” so users can see what the system is doing.

	5V Relay Module	Acts as a switch controlled by the Arduino to turn higher-power devices, like the solenoid lock, on or off safely.
	12V Solenoid Door Lock	The locking mechanism that physically locks or unlocks the door when triggered by the Arduino.
	12V Power Adapter	Provides the necessary power to run the solenoid lock and other 12V components in the system.
	Breadboard & Jumper Wires	Used to connect all components temporarily for testing and assembly without soldering.

Buying the parts online allowed us to get everything we needed conveniently and on time. This phase made sure all materials were ready for building and testing the system, so the team could move forward without delays.

Phase 4 – Component Testing and Assembly

During the 9th and 12th of February, the team is assembling the system and checking that all hardware components are correctly connected. The RFID module, solenoid lock, relay, LCD display, and Arduino board are being set up on the breadboard to ensure proper placement and connections.



At this stage, the team is focused on the physical assembly and verifying that the components are compatible and positioned correctly, preparing the system for coding and full integration in the next phase.

III. Learning Outcomes and Challenges

Learning Outcomes

Through the development of the RFID Vault Access System, the team gained practical experience in assembling Arduino circuits, handling electronic components, and understanding how RFID technology works. Members improved their skills in planning and organizing project tasks, coordinating as a team, and documenting progress clearly. The project also strengthened problem-solving abilities, as the team had to think critically about how to design a secure and functional access control system. Overall, the experience provided both technical knowledge and collaborative skills that will be valuable for future projects.

Challenges Encountered

During the project, the team faced several challenges, mainly related to hardware assembly and component compatibility. Some sensors and modules required careful handling to fit correctly on the Arduino board, and minor connection errors occurred during assembly. Coordinating schedules among group members also slowed down progress at times. Additionally, testing individual components without code presented difficulties in understanding how everything would work together in the final system. These challenges required patience, careful planning, and teamwork to resolve effectively.

Future Directions

The next steps for the project include completing the full assembly of the system, integrating all components, and implementing the Arduino code to enable functional access control. The team also plans to test the system thoroughly to ensure that RFID authentication, access logging, and alert notifications work reliably. In the future, the project could be expanded to include features such as remote monitoring, mobile notifications, or integration with other security systems, further enhancing convenience and security.

IV. GitHub code for the Arduino

```
#include <LiquidCrystal_I2C.h>
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 10
#define RST_PIN 9
String UID = "83 B9 BE 18";
byte lock = 1;

const int RELAY_PIN = 7;

LiquidCrystal_I2C lcd(0x27, 16, 2);
MFRC522 rfid(SS_PIN, RST_PIN);


void setup() {
  Serial.begin(9600);
   pinMode(RELAY_PIN, OUTPUT);

  lcd.init();
```

```
  lcd.backlight();

  SPI.begin();
  rfid.PCD_Init();
}

void loop() {
  lcd.setCursor(4, 0);
  lcd.print("Welcome!");
  lcd.setCursor(1, 1);
  lcd.print("Put your card");

  if ( ! rfid.PICC_IsNewCardPresent())
    return;
  if ( ! rfid.PICC_ReadCardSerial())
    return;

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Scanning");
  Serial.print("NUID tag is :");
  String ID = "";
  for (byte i = 0; i < rfid.uid.size; i++) {
    lcd.print(".");
    ID.concat(String(rfid.uid.uidByte[i] < 0x10 ? " 0" : " "));
    ID.concat(String(rfid.uid.uidByte[i], HEX));
    delay(300);
  }
  ID.toUpperCase();

  if (ID.substring(1) == UID && lock == 0 ) {

    digitalWrite(RELAY_PIN,LOW);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Door is locked");
    delay(1500);
    lcd.clear();
    lock = 1;
  } else if (ID.substring(1) == UID && lock == 1 ) {

    digitalWrite(RELAY_PIN,HIGH);
    lcd.clear();
    lcd.setCursor(0, 0);
```

```
    lcd.print("Door is open");
    delay(1500);
    lcd.clear();
    lock = 0;
  } else {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Wrong card!");
    delay(1500);
    lcd.clear();
  }
}
```

Repository Link:
https://github.com/jshijk/Smart-Security-with-Arduino--Automated-Locking-system

This code is for the Arduino-based RFID door lock system that uses an LCD screen, an RFID reader (MFRC522), and a relay to control a door. When the system starts, it initializes the LCD, the RFID reader, and sets the relay pin as an output. The screen shows a welcome message asking the user to scan their card. When a card is placed near the reader, the system reads the card's unique ID (UID) and compares it to a saved authorized UID. If the scanned UID matches the saved one, the relay switches state to either lock or unlock the door, and the LCD displays the door status. If the card does not match, the screen shows "Wrong card!" and the door state does not change.